HAEBOT / KIRO V2-A ANKER SOLIX C300 DC POWER SYSTEM

Electrical schematic, Rev B, 2026-11-13. STATUS: PROPOSED - build only after the bench checks in docs/verification_report.md

BRANCH D - WHEEL DRIVE: USB-C2 -> 20 V PD trigger -> 12 V buck -> F1 -> K1 (E-stop) -> wheel adapter -> 4 x ST3215-C018



BRANCH C - ARM 2 (powered follower): USB-C1 -> 20 V PD trigger -> EA75-5V buck -> F2 -> K1 (E-stop) -> arm adapter

E-STOP CONTROL (hardware only, no software): DRIVE_12V -> F3 -> S1 (NC) -> START || K1 aux -> K1 coil. Opens wheels, arm 1 and arm 2. Jetson + hubs stay on.

BRANCH B - ARM 1 (leader / torque-off arm only): USB-C4 5 V 3 A -> USB-C 5 V lead -> F4 -> K1 (E-stop) -> arm adapter

BRANCH E - JETSON: car socket 12 V 10 A -> fused plug lead -> F5 -> DC jack 5.5 x 2.5 mm centre + (devkit jack accepts 9-20 V; no boost)

BRANCH A - USB HUBS: USB-A1 / USB-A2 (5 V 2.4 A each) -> DC lead -> hub power input. BLOCKED: hub input voltage unknown

ANKER PORT LIMITS (manual A1726)

USB-C1: 5/9/15 V 3 A, 20 V 5 A (100 W)
USB-C2/C3: + 28 V 5 A (140 W)
USB-C4: 5 V 3 A (15 W)
USB-A1/A2: 5 V 2.4 A (12 W each)
USB-A1 + A2 + C4 together: 20 W
Car socket: 12 V 10 A (120 W)
Total output: 300 W
NO 12 V USB-C profile, no PPS listed

PORT MIX USED ON THIS SHEET

C1 + C2 + (A1,A2,C4) + car socket
= manual 4-port case:
C1 30 W | C2 100 W | USB group 20 W
| car 120 W

CALCULATED MARGINS

Wheels: 78 W design < 100 W OK
Wheels: 141 W all-stall > 100 W
Arm 2: 28 W design < 30 W marginal
Arm 2: 44 W burst > 30 W
Jetson: 45 W peak < 120 W OK
Hubs + Arm 1: must stay < 20 W

WARNINGS

1. Never join the + of two ports.
2. Plugging or unplugging any port can renegotiate PD and drop the other outputs for a moment.
3. Set and measure U3 (12.0 V) and U4 (5.0-5.2 V) before any servo is connected.
4. Negatives of all ports become common through USB shields. Motor current must return on its own pair, not through USB.
5. E-stop cuts wheels + both arms.

STATUS WORDS

EXISTING = part on the V1 robot
PROPOSED = not bought, not tested
BLOCKED = needs a hardware fact

Drawn 2026-11-13. KiCad ERC must be run in KiCad 8+ before release
Single lines to servos = 3-wire Molex 5264 bus cable (GND / VCC / SIG)
Port limits: Anker A1726 user manual. Loads: V1 Rev F records (kiro-wiring/docs)
Status: PROPOSED - not bench-verified. See docs/verification_report.md
Genie EduTech

Sheet: /
File: HAEBOT_ANKER.kicad_sch

Title: HAEBOT / KIRO V2-A - Anker SOLIX C300 DC power system

Size: A3	Date: 2026-11-13	**Rev: B**
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